INCH-POUND

MIL-W-24754(SH)
6 November 1990

MILITARY SPECIFICATION

WASHER, HIGH PRESSURE, ELECTRIC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a portable, high pressure (maximum 800 pounds per square inch (lb/in^2), hot or cold, fresh water washer system, electric.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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SPECIFICATIONS

MILITARY

MIL-S-12651	Sprayers, Packaging of.

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR

Code of Federal Regulations Title 29, Part 1910	Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F1166	Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS-1	General Standards for Industrial Controls and Systems
MG 1	Motors and Generators
WD 1	General Requirements for Wiring Devices

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order or precedence.** In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 **General.** The washer system shall consist essentially of an electric pump; chemical storage and dispensing system; wand assembly; operational and safety controls; and carriage mounting for mobility.

3.3 **Standard commercial product.** The washer shall be in accordance with the requirements of this specification and shall be the contractor's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the contractor's standard commercial product shall be included in the washer being furnished.

3.4 **Materials.** Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials shall be of the same quality used for the intended purpose in commercial practice. Parts which are subject to corrosion in a marine environment shall be either flame sprayed, zinc galvanized, or constructed of stainless steel, brass, bronze, or other corrosion resistant material.

3.4.1 **Recovered materials.** Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.5 Construction.

3.5.1 Human factors engineering. Human factors engineering criteria, principles and practices, as specified in ASTM F1116, shall be used in the construction of the electric, high pressure washer where specified. The equipment shall be constructed so that operation and access for maintenance, and repair shall meet the applicable requirements of ASTM F1116. Access to each control, valve, or any other component shall include adequate working space for the use of any tools, test equipment, and any protective clothing likely to be worn in order to perform the required maintenance task. Maintenance and operational activities shall permit performance by the 5th percentile female to the 95th percentile male as defined in ASTM F1116. Particular attention shall be given to the ease of access for the replacement or cleaning of tank filters. Controls shall be configured in accordance with the applicable requirements of ASTM F1116. Skill levels and training requirements for operation, maintenance and repair as influenced by the machine design shall be kept at the lowest level consistent with other requirements herein. The subsequent design for maintenance and repair functions shall reflect considerations of safety to prevent conditions hazardous to personnel or deleterious to equipment.

3.5.2 Power unit. The motor shall be constructed and rated in accordance with NEMA MG 1. Motor shall be furnished in a dripproof enclosure and sealed windings. Motor controller shall meet the design and operational requirements of NEMA ICS-1.

3.5.3 Pump. The pump shall be driven by a direct coupled electric motor and shall deliver fresh water at the specified volume and pressure. The pump shall be furnished with a pressure gauge and a drainage system.

3.5.4 Valves. A pressure relief, back flow preventer and vacuum breaker valve shall be provided with the unit. The pressure relief (dump) valve shall be located downstream of the pump and upstream of the spray wand. The relief pressure shall not exceed 110 percent of rated discharge. The over pressured discharge shall be routed to the supply side of the pump. The back flow (check) valve shall be located between the supply side of the pump and the water supply line. The vacuum breaker valve shall be located between the check valve and the water supply line.

3.5.5 Wand. The spray wand shall consist of a gun, with trigger and internal parts, high pressure lance and chemical dispensing tube. The wand shall be constructed so that the operator is able to change the nozzle pressure from low pressure chemical dispensing to high pressure rinsing without returning to the machine.

3.5.5.1 Gun. A pistol grip gun shall be equipped with a trigger that controls discharge to the lance. There shall be a trigger guard and an automatic safety lock-out, in the off position, to prevent accidental discharge of high pressure water.

3.5.5.2 Lance. The lance shall be approximately 3 feet long, fabricated from schedule 80 stainless steel. An insulated support sleeve or handle shall protect operating personnel from burns. The high pressure lance shall be equipped with nozzles of three different spray patterns; 15 degrees, 25 degrees, and 45 degrees. The nozzles may be independent or turret type, and shall be corrosion resistant and attached to the high pressure lance by a positive locking method.

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3.5.6 Hoses. The hose shall be the lightest possible high pressure metal braided rubber hose that withstands a working pressure of at least 800 lb/in^2, with a burst pressure of at least 3,200 lb/in^2 and an operating temperature of 150 degrees Fahrenheit (°F). Hose sections shall be provided in 30-foot lengths. A minimum of 60 feet of hose shall be furnished with each unit. A manual takeup reel or other means of storage shall be provided for the hose. A 50-foot long, 3/4-inch inside diameter (id) garden hose or equivalent shall be used for feedwater supply to the washer.

3.5.7 Chemical dispensing system. The chemical dispensing system shall inject chemicals, from a container, downstream of the pump. Control valves shall be provided on both the unit and wand assembly. The chemical container shall be a 1 gallon polycarbonate jug, either transparent or translucent, and shall be secured to the unit carriage. The fluid line leading from the jug shall include a permanent cleanable filter.

3.5.8 Carriage. A two-wheel single axle carriage shall be provided for mobility.

3.6 Electrical system. The electrical system shall be comprised of motors, selector switches, sensors, and wiring hardware to operate with a 115-volt(V), 3-phase, 60 hertz (Hz) power source.

3.6.1 Switch. The electric motor shall be provided with a waterproof start/stop switch.

3.6.2 Grounding. The washer shall be grounded to avoid shock hazards.

3.6.3 Power cord. A 15-foot hard service, oil resistant, flexible power cord shall be hard wired to the ground fault interrupter (GFI). The cord shall be 12 gauge, 3 wire including ground, 600 V NEMA type "SO" with molded plug in accordance with NEMA 5-15P (15-ampere 120 V), conforming to NEMA WD 1.

3.7 Safety. Mechanical or electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded (see 3.12.3).

3.7.1 Safety controls. The cleaner shall be equipped with all mechanical and electrical controls and devices necessary for safe and efficient operation. Such controls shall include the following:

a. Motor safety shutoff control, with reset button, to shut down motor when overheated.

b. Mechanical or electrical pressure control device to prevent the pressure in the delivery hose from exceeding a predetermined safe limit.

c. Built-in ground fault interrupter circuit to shut down motor if improperly grounded.

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3.8 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.9 Cleaning, treatment, and painting. Surfaces normally painted in commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be flame sprayed and then prepared for treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with contractor's current materials according to contractor's current processes and the total dry film thickness shall be not less than 1.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.10 Identification plate. The contractor shall stamp all necessary data in the blank spaces on the identification plates and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts not less than 1/8 inch in diameter shall be used to affix the plates.

3.11 Instruction plates. When any special or important procedure is to be followed in operating and servicing the equipment, an instruction plate describing such procedure shall be affixed to the unit. Plates shall be of copper or zinc-base-alloy, or aluminum, adequately secured. Lettering shall be black on a light background and sized and arranged as specified in ASTM F1116.

3.12 Performance.

3.12.1 Rated capacity. The washer shall produce a flow of not more than 3 gallons per minute, and a nozzle discharge pressure of not more than 800 lb/in^2 (see 4.7.1).

3.12.2 Continuous performance. The washer shall operate continuously for not less than 48 hours (see 4.7.2).

3.12.3 Safety and health standards. The cleaner shall conform to the Code of Federal Regulations, Title 29, CFR, Chapter XVII, Part 1910 (see 4.7.3).

3.12.4 Shock. The washer wand and delivery hose assembly shall withstand a drop of 6 feet without permanent deformation, breaks, or fractures (see 4.7.4).

3.12.5 Electromagnetic interference control. The washer shall meet the electromagnetic interference control requirements and test limits specified in MIL-STD-461 and MIL-STD-462 (see 4.7.5).

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3.13 Workmanship.

3.13.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which are deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. Bends shall be made by controlled means to ensure uniformity of size and shape.

3.13.2 Bolted connections. Bolt holes shall be accurately punched or drilled and the burrs shall be removed. Washers or lockwashers shall be provided in accordance with commercial practice, and all bolts, nuts, and screws shall be tight.

3.13.3 Riveted connections. Rivet holes shall be accurately punched or drilled and the burrs shall be removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.13.4 Welding. Welding procedures shall be in accordance with nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein.

4.4 First article inspection. The first article inspection shall be performed on one complete cleaner. This inspection shall include the examination of 4.6 and the tests of 4.7. The first article may be either a first production item or a standard production item from the contractor's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed as specified herein. This inspection shall include the examination of 4.6 and the tests of 4.7.

4.6 Examination. The washer shall be examined for conformance to the requirements specified herein. Any redesign or modification of the contractor's standard product to conform to specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention. This element of inspection shall include all visual examinations and dimensional measurements. Nonconformance to any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.7 Tests. Failure of any unit to pass any phase of any test shall be cause for rejection.

4.7.1 Performance. The washer shall be connected and operated in accordance with the contractor's operating instructions. With the trigger depressed, the washer shall be operated at the rated discharge of not more than 3 gallons per minute at 800 lb/in^2 for a period of 30 minutes. During this 30-minute period, the motor and pump shall be operated continuously, except that after each 5 minutes of operation, the trigger shall be released and the operation of the pressure reduction system observed for 30 seconds. Pressure shall be reduced to that of the inlet pressure of the pump. Upon completion of this test, no component of the washer (motor, pump, hose connections, control mechanism, and so forth) shall show evidence of failure, damage, or leakage.

4.7.2 Continuous operation. This test shall be performed at an ambient temperature of 75 $\pm$ 5 °F. With the trigger depressed, the washer shall be operated at a rated discharge of 3 gallons per minute at 800 lb/in^2 for 48 hours. After each 30 minutes of operation, the trigger shall be released and operation of the pressure reduction system shall be observed for 30 seconds. Pressure shall be reduced to that of the inlet pressure of the pump. As a minimum, 8 full cycles per day shall be performed until the machine has been operational for a period of 48 hours. Upon completion of this test, there shall be no evidence of failure of, damage to, or leakage by any component of the washer (motor, pump, hose connection, control mechanisms, and so forth).

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4.7.3 **Safety control.** After the test of 4.7.2, the washer shall be tested to verify conformance to 3.7. The conditions which activate the safety controls shall be simulated by interruption of water flow, and for the pressure control test, by throttling the discharge.

4.7.4 **Drop.** The washer wand and delivery hose assembly shall be dropped two times from a height not less than 6 feet onto a smooth concrete floor. One drop shall impact on the handle and one drop shall impact on the wand. Upon completion of the two drops, the handles and wand shall be inspected for permanent deformation, breaks, and fractures. Either breakage or fracture of the handle and wand shall be cause for rejection. Permanent deformation of the handle and wand shall not be cause for rejection. The washer shall then be operated, and the handles and wand inspected for leaks. Leaks correctable by tightening of threaded connections shall not be cause for rejection. Leaks that cannot be corrected by tightening of threaded connections shall be cause for rejection. Failure of the cleaner to operate satisfactorily shall be cause for rejection.

4.7.5 **Electromagnetic interference control.** The first article shall be tested to meet the requirements of MIL-STD-461, UM05 for group I equipment.

4.8 **Inspection of packaging.** Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 **Preservation, packing and marking.** Preservation, packing and marking shall be in accordance with the requirements of MIL-S-12651 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 **Intended use.** The equipment covered by this specification is intended for use in heavy duty shipboard cleaning tasks such as: bilges, interior bulkheads, heads, showers, galleys, scullerics and medical spaces. This equipment should be operated by two men, one operating the pump, the other operating the wand.

6.2 **Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Whether first article inspection is required (see 3.1)
c. Color of finish coat required (see 3.9)

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d. Test conditions, if other than as specified (see 4.3)
e. Level of preservation and packing required (see 5.1).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DOD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), and included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (key word) listing.

Delivery hose
Lance
Wand

Preparing activity:
Navy - SH
(Project 4940-N564)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-24754(SH)	2. DOCUMENT DATE (YYMMDD) 6 November 1990

3. DOCUMENT TITLE
WASHER, HIGH PRESSURE, ELECTRIC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME NAVSSES 074E Mr. D.C. Beeman (TPC)	b. TELEPHONE (Include Area Code) (1) Commercial: 443-7309 (2) AUTOVON: 8-443-7309
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command 5523 Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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